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Name _____

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School/Department _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} e^{-t} t^{\nu-1} dt, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}; \quad \int_{-\infty}^{+\infty} e^{-x^2} dx = \sqrt{\pi}.$$

c	h	$\hbar$	e	m_e	m_n	u	N_A	k_B
3.0×10^8 m/s	6.63×10^{-34} J·s 1.24×10^3 eV·nm/c	1.05×10^{-34} J·s	1.6×10^{-19} C	9.11×10^{-31} kg $0.511 \text{ MeV}/c^2$	939.565 MeV/c^2	931.494 MeV/c^2	6.02×10^{23} mol^{-1}	8.617×10^{-5} eV/K

Quantum numbers of three quarks:

Quark symbol	Charge number	Spin	Baryon number	Strangeness
u	2/3	1/2	1/3	0
d	-1/3	1/2	1/3	0
s	-1/3	1/2	1/3	-1

Select five out of the following six problems.

1.(20 pts.) In the experiment of photoelectric effect using iron as emitter, it is found that the threshold frequency of light is $\nu_C = 1.10 \times 10^{15}$ Hz.

(a) Find the work function ϕ of iron in the unit of eV.

(b) Find the stopping voltage V_0 for the photoelectrons emitted when light of wave length $\lambda = 250 \text{ nm}$ falls on the surface of iron.

2. (20 pts.) In the beta decay of ${}^{24}_{11}\text{Na}$, the daughter nucleus is ${}^{24}_{12}\text{Mg}$ and an electron with kinetic energy of 2.15 MeV is observed. The atomic mass of the ${}^{24}_{11}\text{Na}$ is $m({}^{24}_{11}\text{Na}) = 23.990963 \text{ u}$ and the atomic mass of ${}^{24}_{12}\text{Mg}$ is $m({}^{24}_{12}\text{Mg}) = 23.985042 \text{ u}$.

(a) Write down the nuclear equation of the decay.

(b) Find the Q-value of the decay.

(c) What is the energy of the accompanying neutrino/antineutrino?

3. (20 pts.) For each of the following reactions, identify it is allowed or forbidden. If it is forbidden, state a conservation law that is violated.

(a) $p + p \rightarrow p + \Lambda^0 + K^+$

(b) $K^0 \rightarrow \pi^+ + e^-$

(c) $p + n \rightarrow p + \Lambda^0$

(d) $\Omega^- \rightarrow \Lambda^0 + K^-$

(e) $n + \nu_e \rightarrow p + e^-$

(f) $\Lambda^0 \rightarrow p + \pi^-$

(g) $p + \bar{p} \rightarrow \Lambda^0 + \Lambda^0$

The quark combinations of relevant hadrons:

$$p - uud \quad \Lambda^0 - uds \quad \Omega^- - sss \quad \bar{p} - \bar{u}\bar{u}\bar{d} \quad K^+ - u\bar{s} \quad K^0 - d\bar{s} \quad K^- - s\bar{u} \quad \pi^+ - u\bar{d} \quad \pi^- - d\bar{u}$$

4. (20 pts.) An ideal Bose or Fermi gas is composed of particles whose energy-momentum relation is $\varepsilon = p^2/(2m)$. The distribution function, i.e., the average number of particle in a quantum state with energy ε , can be written as $f(\varepsilon) = 1/(e^{\beta(\varepsilon-\mu)} \pm 1)$ where $\beta = 1/k_B T$, μ is the chemical potential of the gas, and the positive and negative signs are for Fermi and Bose gas respectively. However, if $e^{-\beta\mu} \gg 1$, the ± 1 in the denominator is negligible and the quantum distributions both reduce to the classical limit of Boltzmann distribution $f_C(\varepsilon) = e^{-\beta(\varepsilon-\mu)}$.

(1) Show that the density of states (DOS) of the gas can be expressed as

$$\rho(\varepsilon) = \frac{g_s V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \varepsilon^{1/2}$$

where g_s is the degree of degeneracy due to the spin of the particle and V is the volume of the gas.

(2) Show that total number of particles N in the classical Boltzmann limit can be expressed as

$$N = g_s V e^{\beta\mu} \left(\frac{2\pi m}{\beta h^2} \right)^{3/2}.$$

(3) Furthermore, show that the condition for the classical limit (or the non-degenerate limit) $e^{-\beta\mu} \gg 1$ is equivalent to the condition $n\lambda^3 \ll g_s$ where $n = N/V$ is the number density of particles and $\lambda = h/\sqrt{2\pi m k_B T}$ is the thermal de Broglie wavelength.

5. (20 pts.) An electron of mass m moves in a one-dimensional symmetric infinite potential well

$$U(x) = \begin{cases} 0, & |x| \leq a/2, \\ +\infty, & |x| > a/2, \end{cases}$$

where $a > 0$ is the width of the well.

(a) Write down the stationary Schrödinger equation of the electron.

(b) Assuming that the solutions of the stationary Schrödinger equation take the form

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ika}, & |x| \leq a/2, \\ 0, & \text{elsewhere} \end{cases}, \quad k > 0, \quad A > 0,$$

determine the possible values of wave vector k by applying the proper boundary conditions.

(c) Using the results of (b) to find the energy levels of the electron in the potential well.

(d) Determine the coefficients A and B in (b) by the normalization of the wave functions.

6.(20 pts.) The Hamiltonian operator of a one-dimensional harmonic oscillator is $\hat{H} = \hat{p}^2 / (2m) + (1/2)m\omega^2 x^2$ where m is the mass of the particle, $\hat{p} = -i\hbar(d/dx)$ is the momentum operator and ω is the natural circular frequency of the oscillator. The normalized ground state wave function of the oscillator is found to be

$$\psi_0(x) = \left(\frac{\alpha}{\sqrt{\pi}} \right)^{1/2} e^{-\frac{\alpha^2 x^2}{2}}, \quad \alpha = \left(\frac{m\omega}{\hbar} \right)^{1/2}.$$

- Find the expectation values of position coordinator $\langle x \rangle$ and momentum $\langle p \rangle$ of the particle in the ground state.
- Find the uncertainties of position $\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}$ and momentum $\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2}$, and show that the Heisenberg's uncertainty principle is satisfied.
- Find the expectation values of kinetic energy and potential energy of the oscillator in its ground state. Comment on their non-zero values and relative magnitude of two quantities.